Segment-4 guidelines al questions answers are given here

Segment 04 — Induction Motor (IM)

Answers compiled from the course slides (Segment 04)

Q1. 3-Phase Induction Motor — Construction, Working Principle, and Equivalent Circuit

1. Introduction

- An induction motor is also known as the asynchronous motor.
- It is basically an AC motor that runs at a speed lower than the synchronous speed.
- It operates on the principle of electromagnetic induction.
- It can run either as an induction motor or as an induction generator.
- Synchronous speed (Ns) is the speed at which the stator's rotating magnetic field turns; it is directly proportional to the supply frequency: Ns = 120f/P.
- Traditionally used for constant-speed service, but can be used for variable-speed applications with a Variable Frequency Drive (VFD).
- Induction motors are simple, rugged, low-priced, easy to maintain, and best suited for industrial requirements.

2. Construction

Stator:

- Consists of a steel frame enclosing a hollow, cylindrical core made of thin laminations of silicon steel.
- A number of evenly spaced slots are provided on the inner periphery of the laminations to hold the 3-phase winding.

Rotor:

- Mounted on a shaft; a hollow laminated core with slots on its outer periphery.
- The rotor winding placed in these slots is of two types: (i) Squirrel cage type (ii) Wound type.

Squirrel Cage Rotor:

- A laminated cylindrical core with parallel slots on its outer periphery.
- Each slot holds an aluminium or copper bar; the bars are short-circuited at both ends by metal end-rings.
- This entire structure forms a 'squirrel cage', hence the name.
- As the rotor bars are permanently short-circuited, this rotor has low starting torque.

Wound-Type Rotor:

- A cylindrical laminated core carrying a 3-phase winding.
- The open ends of the rotor winding are connected to 3 insulated slip rings on the shaft, with carbon brushes resting on them; the brushes connect to an external 3-phase rheostat.
- At starting, external resistance is added in the rotor circuit to give a large starting torque.
- The rheostat resistance is gradually reduced to zero as the motor comes up to normal running speed.

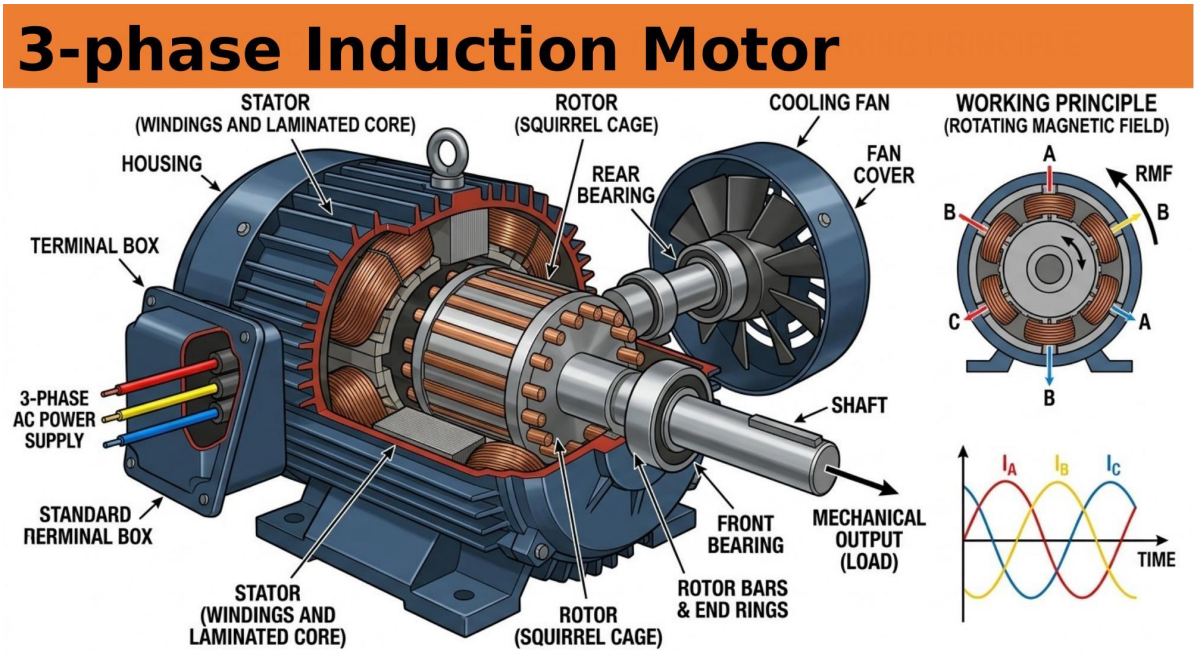


Figure: Construction of a 3-phase induction motor (stator, squirrel-cage rotor, cooling fan, and working principle).

3. Working Principle (Principle of Operation)

- When the 3-phase stator winding is energised from a 3-phase supply, a rotating magnetic field (RMF) is set up which rotates around the stator at synchronous speed Ns.
- This rotating field passes through the air gap and cuts the (stationary) rotor conductors. Because of the relative speed between the rotating flux and the stationary rotor, an e.m.f. is induced in the rotor conductors (Faraday's law of electromagnetic induction).
- Since the rotor circuit is short-circuited (squirrel cage) or closed through external resistance (wound rotor), this induced e.m.f. drives a current through the rotor conductors.
- These current-carrying rotor conductors, lying in the stator's magnetic field, experience a mechanical force (motoring action).
- The sum of the forces on all rotor conductors produces a net torque that drives the rotor in the same direction as the rotating field.
- The rotor tries to catch up with the RMF speed but can never actually reach it — if it did, there would be no relative speed, no induced e.m.f., no rotor current, and hence no torque. This is why the machine is called 'asynchronous'.
- Rotating Magnetic Field: when the 3-phase winding is energised, the field's poles do not stay fixed but keep shifting position around the stator; the speed of this shift is the synchronous speed (Ns).
- Effect of Slip: the difference between the synchronous speed (Ns) of the rotating field and the actual rotor speed (N) is called the slip, usually expressed as a percentage of Ns: $s = (N_s - N)/N_s$.

4. Equivalent Circuit

Note: The Segment-04 slide deck does not include a separate labelled 'equivalent circuit' figure — the construction/working-principle diagram above (Figure, Slide 6) is the only circuit-related figure provided. For completeness, the standard per-phase equivalent circuit of a 3-phase induction motor (as normally taught alongside this topic) is summarised below:

- The stator side is represented by the stator resistance R1 and stator leakage reactance X1, in series with the supply voltage.

- The magnetising branch (core-loss resistance R0 in parallel with magnetising reactance X0) is connected across the stator, similar to a transformer's equivalent circuit.
- The rotor side (referred to the stator) is represented by rotor resistance R2' and rotor leakage reactance X2', in series with the induced e.m.f.
- Because the rotor rotates, its effective resistance is split as R2'/s, which can be written as R2' + R2'(1−s)/s — the R2'(1−s)/s term represents the mechanical (developed) power, i.e. the load carried by the motor.
- This is analogous to a transformer's equivalent circuit, with the 'load resistance' term R2'(1−s)/s standing in for the mechanical output.

Q2. Why is a 1-Phase Induction Motor Not Self-Starting? — Working & Starting Mechanism

1. Construction

- A single-phase induction motor is very similar to a 3-phase squirrel-cage induction motor.
- It consists of a squirrel-cage rotor identical to that of a 3-phase motor, and a single-phase winding on the stator.

2. Why It Is Not Self-Starting (Double-Field Revolving Theory)

- A single-phase induction motor is not self-starting.
- When the rotor is stationary and the single stator winding is connected to a single-phase AC supply, the winding produces a magnetic field that simply pulsates (grows and shrinks) sinusoidally along one axis — it does not rotate.
- This pulsating flux can be resolved (Double Field Revolving Theory) into two rotating fluxes, each of half the maximum magnitude, rotating at synchronous speed in opposite directions — one clockwise and one anticlockwise.
- Each of these two fluxes tends to produce its own induction-motor torque: one torque acts clockwise, the other anticlockwise.
- At standstill, these two torques are exactly equal in magnitude and opposite in direction, so the net starting torque is zero — hence the motor cannot start on its own.
- (Once the rotor is given an initial push in either direction by some external means, the balance is broken, one field's torque dominates, and the motor will continue to accelerate and run in that direction.)

3. Starting Mechanism

- To make the motor self-starting, an additional stator winding — the start/auxiliary winding — is added, displaced from the main winding.
- The start winding may include a series capacitor and/or a centrifugal switch.
- When supply voltage is applied, current in the main winding lags the supply voltage (due to the main winding's impedance).
- At the same time, current in the start winding leads or lags the supply voltage depending on the starting-mechanism impedance (e.g. capacitor), so the two currents are out of phase with each other.
- The interaction between the magnetic fields of the main winding and the start winding produces a resultant rotating magnetic field, and the motor starts rotating in the direction of this resultant field.
- Once the motor reaches about 75% of its rated speed, a centrifugal switch disconnects the start winding.
- From that point on, the single main winding alone is able to maintain sufficient torque (via the dominant rotating field, per the double-revolving-field theory) to keep the motor running on its own.

Single-Phase Induction Motors

- The starting mechanism of the single-phase induction motor is mainly an additional stator winding (start/ auxiliary winding) as shown in Figure. The start winding can have a series capacitor and/or a centrifugal switch. When the supply voltage is applied, current in the main winding lags the supply voltage due to the main winding impedance. At the same time, current in the start winding leads/lags the supply voltage depending on the starting mechanism impedance. Interaction between magnetic fields generated by the main winding and the starting mechanism generates a resultant magnetic field rotating in one direction. The motor starts rotating in the direction of the resultant magnetic field.
- Once the motor reaches about 75% of its rated speed, a centrifugal switch disconnects the start winding. From this point on, the single-phase motor can maintain sufficient torque to operate on its own.

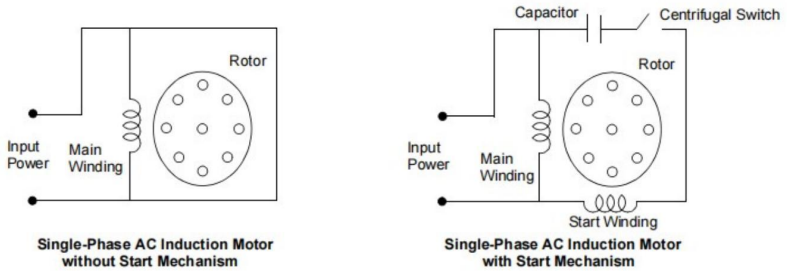


Figure: Single-phase induction motor without a start mechanism vs. with a start (auxiliary) winding and centrifugal switch.

Q3. Numerical Problems — Examples 34.3, 34.4, 34.5

Example 34.3

A 4-pole, 3-phase induction motor operates from a supply whose frequency is 50 Hz. Calculate: (i) the speed at which the magnetic field of the stator is rotating, (ii) the speed of the rotor when the slip is 0.04, (iii) the frequency of the rotor currents when the slip is 0.03, (iv) the frequency of the rotor currents at standstill.

Solution:

(i) Synchronous speed: $N_s = 120f/P = (120 \times 50)/4 = 1500$ r.p.m.

(ii) Rotor speed: $N = N_s(1 - s) = 1500(1 - 0.04) = 1440$ r.p.m.

(iii) Rotor current frequency: $f' = s \times f = 0.03 \times 50 = 1.5$ Hz (= 90 cycles/min)

(iv) At standstill, $s = 1$, so $f' = s \times f = 1 \times 50 = 50$ Hz

Example 34.4

A 3-phase induction motor is wound for 4 poles and is supplied from a 50-Hz system. Calculate (i) the synchronous speed, (ii) the rotor speed when slip is 4%, and (iii) the rotor frequency when the rotor runs at 600 r.p.m.

Solution:

(i) $N_s = 120f/P = (120 \times 50)/4 = 1500$ r.p.m.

(ii) $N = N_s(1 - s) = 1500(1 - 0.04) = 1440$ r.p.m.

(iii) When $N = 600$ r.p.m.: $s = (N_s - N)/N_s = (1500 - 600)/1500 = 0.6$

Rotor current frequency: $f' = s \times f = 0.6 \times 50 = 30$ Hz

Example 34.5

A 12-pole, 3-phase alternator driven at a speed of 500 r.p.m. supplies power to an 8-pole, 3-phase induction motor. If the slip of the motor, at full load, is 3%, calculate the full-load speed of the motor.

Solution:

Supply frequency generated by the alternator: $f = (P_{alt} \times N_{alt})/120 = (12 \times 500)/120 = 50$ Hz

Synchronous speed of the motor: $N_s = 120f/P = (120 \times 50)/8 = 750$ r.p.m.

% slip: $s = (N_s - N)/N_s \times 100 \Rightarrow 3 = (750 - N)/750 \times 100$

Solving: $N = 750 - (3 \times 750/100) = 750 - 22.5 = 727.5$ r.p.m.

∴ Full-load speed of the motor, N = 727.5 r.p.m.

Q4. Working of a Variable Frequency Drive (VFD) — How It Controls Motor Speed

1. What Is a VFD?

- A VFD is a system built from power-electronics devices (IGBTs, MOSFETs, etc.), a high-speed central controller (a microcontroller such as a PIC18/PIC16/PIC32), and optional sensing devices, depending on the application.
- Its basic function is to act as a variable-frequency generator, so that it can vary the speed of the motor according to the user's setting.
- Since the induction motor's synchronous speed is Ns = 120f/P, changing the supply frequency f (with pole number P fixed) directly changes Ns, and hence the actual running speed N = Ns(1 − s). This is the fundamental principle behind VFD speed control.

2. Why Is a VFD Important?

- Avoids over-design: Without a drive, motors are often oversized to handle peak loads, wasting energy during normal operation.
- Variable speed control: For loads like fans and pumps, power consumption is proportional to the cube of speed, so a small speed reduction can cut power usage by up to 50%.
- Eliminates throttling: Instead of mechanical valves that waste energy as friction/heat, the drive simply slows the motor down to reduce flow.
- Soft starting: Induction motors draw large inrush currents at startup (causing voltage dips and heat stress); a VFD provides a controlled ramp-up, protecting the motor insulation and the local grid.
- Power-factor correction: At light loads, induction motors have very poor power factor (around 0.1); modern drives help maintain a better PF, avoiding utility penalties.

3. VFD Block Diagram / Working Stages

Variable Frequency Drive (VFD)

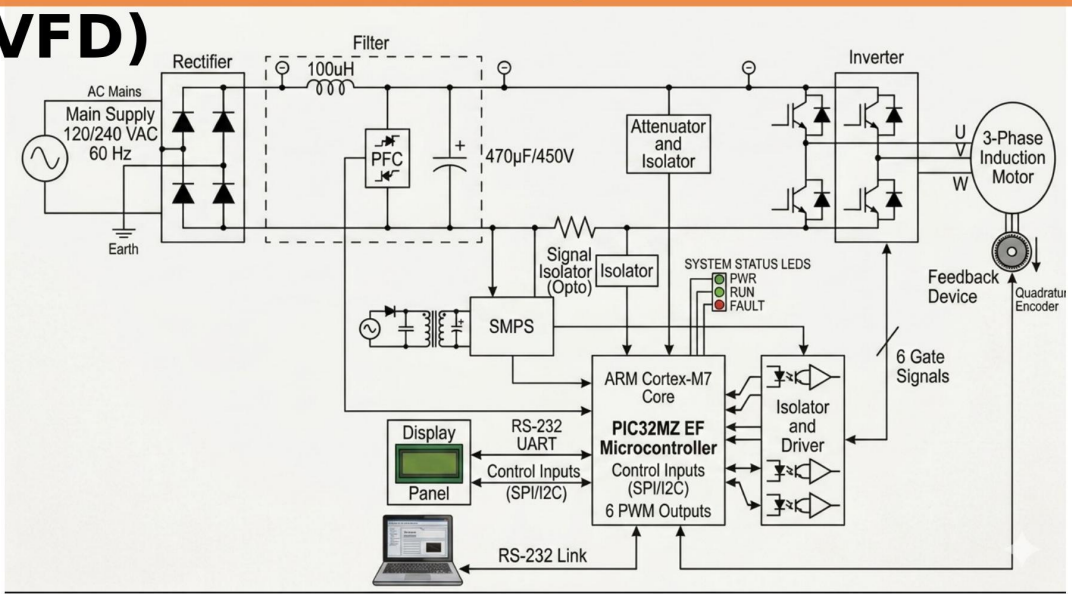


Figure: VFD block diagram — Rectifier → Filter/DC bus → PIC32MZ EF Microcontroller → Gate Isolator/Driver → Inverter → 3-phase Induction Motor, with encoder feedback.

Rectification:

- AC Main Supply provides the primary electrical power (120/240 VAC).
- A bridge rectifier (four diodes) converts the incoming AC into DC.

Filter Section:

- An inductor smooths out current ripples and filters high-frequency noise.
- PFC (Power Factor Correction) improves efficiency by ensuring near-unity power factor.
- A capacitor (470 µF / 450 V) further smooths the DC voltage to give a stable DC bus for the inverter.

Control & Processing:

- The PIC32MZ EF microcontroller (with ARM Cortex-M7 core) is the 'brain' of the system: it processes feedback, executes the control algorithm, and generates 6 PWM (Pulse Width Modulation) outputs to drive the motor.

Motor Drive & Feedback:

- The Isolator and Driver amplifies the low-power PWM signals from the microcontroller to high-power levels able to switch the inverter's transistors, while also providing electrical isolation between the logic and power stages.
- The Inverter uses six IGBTs or MOSFETs to switch the DC bus voltage into a 3-phase AC output. By varying the switching timing (i.e., the output frequency and voltage), it controls the speed and torque of the motor — this is the actual speed-control action.
- A Feedback Device (quadrature encoder) attached to the motor shaft sends real-time position and speed data back to the microcontroller, enabling closed-loop control.

Logic Power & Sensing:

- An SMPS (Switched-Mode Power Supply) steps down the high DC bus voltage to low-voltage DC (e.g. 3.3 V or 5 V) to power the microcontroller.
- A signal isolator (opto-isolator), attenuator, and isolator maintain a safety mechanism so the microcontroller can monitor the bus voltage without being damaged.
- Control inputs (from a control panel) reach the microcontroller via SPI/I2C, and an RS-232/UART link allows external communication.

Variable Frequency Drive (VFD)

Rectification
•**AC Main Supply:** Provides the primary electrical power (120/240 VAC).
•**Rectifier:** Uses a a bridge rectifier configuration using four diodes to convert the incoming AC into DC.

The Filter Section
•**Inductor a**cts to smooth out current ripples and filter out high-frequency noise.
•**PFC (Power Factor Correction):** Improves the efficiency of the power supply by ensuring unity power factor.
•**Capacitor ($470\mu F / 450V$):** further smooths the DC voltage to provide a stable DC supply for the inverter.

Control & Processing
•**PIC32MZ EF Microcontroller :** The "brain" of the system. It processes feedback, executes control algorithms, and generates the **6 PWM (Pulse Width Modulation)** Outputs needed to drive the motor.

Motor Drive & Feedback
•**Isolator and Driver:** Amplifies the low-power PWM signals from the MCU to high-power levels capable of switching the Inverter's transistors. It provides electrical isolation between the logic and power stages.
•**Inverter:** Uses six IGBTs or MOSFETs to switch the DC bus voltage into a 3-phase AC output. By varying the timing, it controls the speed and torque of the motor.
•**Feedback Device (Quadrature Encoder):** Attached to the motor shaft. It sends real-time position and speed data back to the microcontroller to enable "Closed-Loop" control.

Logic Power & Sensing
•**SMPS (Switched-Mode Power Supply) s**teps down the high DC bus voltage to low-voltage DC (e.g., 3.3V or 5V) to power the microcontroller and **Signal Isolator (Opto), Attenuator and Isolator** maintains a safety mechanism so that the microcontroller can monitor the voltage without being damaged.

Figure: Detailed VFD stages — AC supply, rectifier, filter/PFC/capacitor, PIC32MZ EF controller, gate isolator & driver, inverter, encoder feedback, and SMPS/logic power.

4. How the VFD Controls Motor Speed (Summary)

- The incoming AC supply is rectified to DC and filtered/smoothed to form a stable DC bus.
- The microcontroller generates PWM switching patterns that command the inverter's IGBTs/MOSFETs.
- The inverter uses these PWM patterns to synthesise a 3-phase AC output at the desired (variable) frequency and voltage from the DC bus.
- Because the motor's synchronous speed Ns = 120f/P depends directly on the supply frequency f, varying the inverter's output frequency varies the induction motor's running speed.
- The encoder feeds real-time speed/position data back to the microcontroller, which adjusts the PWM pattern continuously — giving accurate, closed-loop speed and torque control.